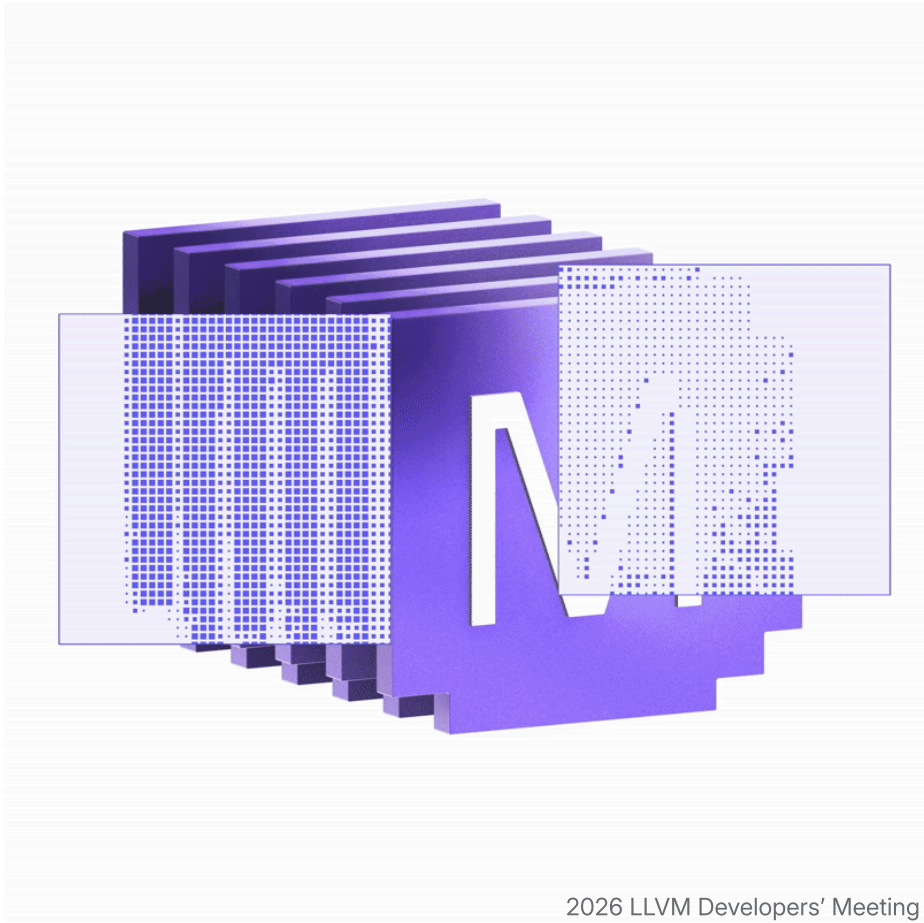
Modular's open-source journey

Joe Loser
Brad Larson



Modular

2026 LLVM Developers' Meeting
Mojo: Internals Deep Dive and Roadmap

01 Three years in the open

02 What’s public today

03 How we ship open source

04 Where you can contribute

What we’ll cover
in the next twenty minutes

Three years
in the open

We opened the stack one layer at a time, from the standard library to the Mojo compiler and the internals of MAX. Our process, tooling and experience improved with each release.

2023

GitHub repos

The modular GitHub repositories are established

MAR 2024

Mojo stdlib

The Mojo standard library is open-sourced

MAY 2025

Kernels

450k+ lines of MAX kernels, plus the rest of the stdlib

NOV 2025

MAX Python APIs

Opened through 2025, last pieces in November

AUG 18 2026

Mojo language

Fully open-sourced at ModCon under Apache 2.0 with LLVM exceptions

OCT 22 2026

MAX internals

The remaining internals of MAX become source-available

Opening the standard library, March 2024: a success disaster.

01

Drowning in contributions.
Two stdlib engineers faced hundreds of new contributors. We needed to hire, and to set clearer expectations about how fast we could review.

02

Infrastructure still maturing.
The tooling to sync external PRs into our internal monorepo wasn't yet built for this scale.

03

A disconnect in direction.
The community pushed Mojo toward a Python superset. Internally we needed it for high-performance CPU and GPU code.

What changed: a public Mojo roadmap, Mojo repositioned as a language in the Python family, and new features proposed in a GitHub issue before a PR. Contributor priorities started matching ours.

Opening the kernels, May 2025: the community and our roadmap lined up.

450k+

Lines of Mojo and MAX code opened in one release

The MAX kernels and the rest of the standard library, including GPU programming. With a public roadmap and Mojo positioned in the Python family, new contributors were working on the same problems we were.

Opening the compiler, August 2026: applying the lessons.

01

Source first.
The compiler, tooling and build source shipped at ModCon under Apache 2.0 with LLVM exceptions.

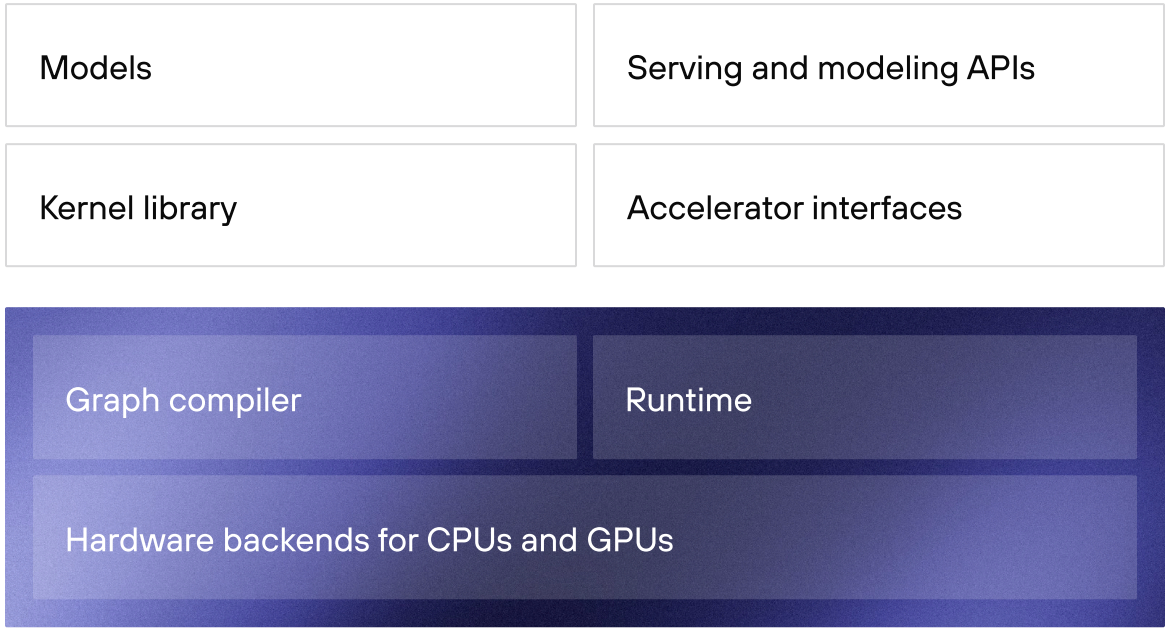
02

Contributions second, in phases.
From Mojo 1.1 on September 17: bug fixes first, then refactorings. New features start as a GitHub issue.

03

Direction up front.
Mojo 1.0's stability guarantee and a public roadmap told contributors where the language is headed.

Opening MAX, October 2026: the whole stack now builds from public source!



APACHE 2.0

Opened through 2025 in Mojo and Python. These stay Apache 2.0.

NEW MAX LICENSE

Source-available October 22, 2026. Free to use and build commercial products on, with no fees or royalties. Open to bug-fix contributions now.

What's public today in github.com/modular/modular

Everything you have seen in today's presentations can be inspected in this repository and tried out at home.





Build it all from source All of Mojo and MAX, in one repository

`github.com/modular/modular`

modular/modular, the monorepo by the numbers

One public repository holds the Mojo compiler, the standard library, the MAX kernel libraries, and the examples. Likely the largest open collection of CPU and GPU kernels anywhere.

30k

GitHub stars

Grown steadily since the stdlib opened in 2024.

416

Contributors

An increasing share from outside Modular.

2,741

Public PRs to date

Every one reviewed in the open, then carried through internal CI.

2,814

Issues closed

889 still open and triaged: bugs, proposals, and questions.

Everything flows
from one monorepo

The internal modularml/modular monorepo is the single source of truth. Every public repository is published from it with Copybara.

SOURCE OF TRUTH

modularml/modular

Everything lives together and is versioned together:

- Compiler and toolchain
- Standard library
- CPU and GPU kernel libraries
- Documentation and examples
- Tests, benchmarks, and CI

COPYBARA

modular/modular	The public monorepo: compiler, stdlib, kernels, examples
modular/vscode-mojo	Language extension for VS Code
modular/mojo-gpu-puzzles	GPU programming puzzles
modular/mojo-quest	Interactive language exercises
modular/max-llm-book	Building LLMs with MAX
modular/skills	Agentic skills for Mojo and MAX
modular/modular-community	Community package registry

Nightly and stable releases, in lockstep across every repository.

All of our open-source repositories update together on a nightly cadence, and only if tests pass.

Nightly
Same commit, every repo

- Every repo advances to the same commit each night
- Not every nightly passes the full test suite
- No green tests, no release that night

Stable
Every ~6 weeks

- Versioned releases cut roughly every six weeks
- MAX branches are calver: `26.1`
- Mojo branches are semver: `1.1`

Docs
Follow the release

- Documentation defaults to stable
- Nightly docs behind a version toggle
- Every version browsable from the picker

How a community PR lands. Reviewed in the open, then tested on a far wider set of systems internally.

01

PR opened

Against public main on modular/modular, which tracks the most recent nightly build.

02

External review

Reviewed in public on GitHub by maintainers and peers.

03

Sync with ！sync

Pulled into the internal monorepo, where CI runs across an expanded set of hardware.

04

Internal review

A second internal review pass, with any modifications made before merge.

05

Merged and closed

The internal merge closes the external PR on the next successful nightly containing the change.

Mojo's new home: mojolang.org

Mojo now has its own site, separate from modular.com. The language, its documentation, and everything you need to learn it, in one place.

`/docs`

Manual and API reference

- Versioned alongside the language
- Stable by default at 1.0
- Nightly in the version picker

`/install`

Quickstart and packages

- Install paths for Linux and macOS
- The community package index
- Verified against every stable release

`/docs/roadmap`

Vision, roadmap, FAQ

- The public phase plan
- Design proposals as they are drafted
- Updated whenever direction changes

`llms.txt`

Built for agents too

- Every page available as Markdown
- Append .md to any URL
- A full index lives at llms.txt

Build Mojo
from source



1. **Clone the public repo.** Mojo lives side by side with the stdlib, MAX, and the rest.
2. **Build the toolchain.** The //Mojo:mojo target builds the full compiler from source.
3. **Test the stdlib** against your locally built compiler.

Build MAX
from source

NEW MAX internals source-available as of October 22, 2026

1. **Use the same clone.** Bazel builds the Mojo compiler, the graph compiler, the kernels, and more.
2. **Run a one-op graph,** the “hello world” of MAX.
3. **Run the kernel tests** on your own GPU.
4. **Serve a full AI model** built entirely from source.



Where you can contribute
to Mojo and MAX

🔥 Mojo contributor guide
`modular/modular › mojo/docs/contributing`

🧑‍🚀 MAX contributor handbook
`modular/modular › max/docs/contributing`



Mojo libraries

The libraries teams post language issues publicly and work in the open. Pick one up.



Mojo compiler

Newly open to bug fixes and more in the core language, compiler, and tooling.



VS Code and LSP

Help improve the VS Code extension and the language server behind it.



MAX

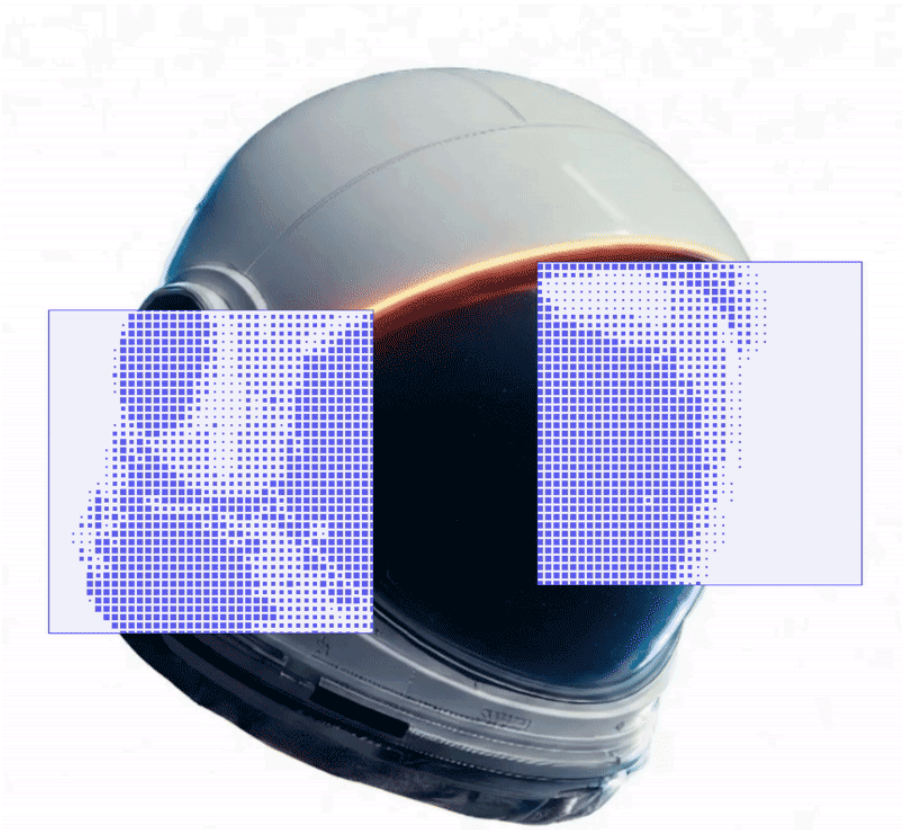
A published roadmap lays out which areas of MAX accept contributions, and when.

Take home: clone the repo. Dig in to the language and frameworks, try them out, and help us push Mojo and MAX forward.

```
$ git clone https://github.com/modular/modular.git
```

Thank you!

github.com/modular/modular
mojolang.org
forum.modular.com



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